

PRIME DNA

The Odd Weil Form on a Window

The sliding-window identity, the logarithmic edge law,
and the bounded-relay conjecture

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Abstract

Let $\lambda(a)$ be the floor of the odd-sector Weil form over test functions supported in $[-a, a]$. Weil's criterion, in Yoshida's odd-test form, says that $\lambda(a) \geq 0$ for every a is equivalent to the Riemann Hypothesis; the floor is certified positive to $a = 0.8$ (Zhu) and decays doubly exponentially in a . We study the floor as a function of the window. Three results are exact. First, the archimedean part of the form on a window is one half of the Dirichlet form of the logarithmic Laplacian plus a bounded form, with an explicit confining wall $\frac{1}{2} \log(1/(a - |y|))$ at the edge; by the boundary theory of the logarithmic Laplacian (Chen–Weth; Hernández-Santamaría, López Ríos, Saldaña) the minimizer vanishes at the edge like $C(a) (\log(a/\delta))^{-1/2}$, and we measure the amplitude $C(a)$ on meshes graded to 10^{-12} of the edge. Consequently the entry of a prime power into the window produces no kink in λ but a soft kink, λ' continuous with an increment $4\Lambda(n)n^{-1/2}C^2/\log(1/\varepsilon)$ at scale ε , which we verify over four decades. Second, between entries the floor obeys the sliding-window identity

$$a \lambda'(a) = - \left[D_\infty(f) + D_P(f) + 2P(f)^2 + \frac{2P(f)}{a} M(f) \right],$$

where f is the minimizer, D_∞ is the archimedean dilation form (its symbol tends to 1), $D_P(f) = -\sum_n 2\Lambda(n)n^{-1/2} \log n g'_f(\log n)$ is the Euler product differentiated against the derivative of the autocorrelation of f at the prime-power positions, and P, M are the polar moments; we verify it to twenty digits in ball arithmetic, exhibiting the cancellation of three order-one quantities to the size of λ . Third, the derivative is a boundary quantity: on eight supports, with and without primes, $\lambda'(a) = -2C(a)^2$ to a part in a thousand, so $\Phi' = 2C^2/\lambda$ and the conjecture below becomes a bound on the edge amplitude of the minimizer by its energy times the horizon. Fourth, the derivative of the floor is a property of the ground state alone: the pencil $(-Q', Q)$ has eigenvalues of size $6 \times 10^{10} T^*$ at $a = 1$ while the minimizer's direction gives $3.7 T^*$, so no inequality between the two forms can hold. With $\Phi = -\log \lambda$ and the horizon $T^*(a) = 2\pi e^{2a}$, a certified computation of the floor from height 17 to height 130 (down to $\lambda = 10^{-93}$, with 230 modes and rigorous eigenvalue enclosures) shows Φ'/T^* rising from 3.2 to 3.9 by shrinking steps on the coarse grids, and the exact derivative resolves a sawtooth around every prime-power entry, a rise of 30–45% into the entry as the form without the entering prime heads toward indefiniteness, then a drop of 9–16% across it, with peaks between 4.1 and 4.3 and dips to 2.3. We state the bounded-relay conjecture $\Phi'(a) \leq c T^*(a)$, which implies the Riemann Hypothesis by integration from Zhu's certified support, and the decay law $\Phi'/T^* \rightarrow c_0$; the data exclude the raw phase-space count, disfavour a logarithmic law, and cannot separate a constant from Zhu's

Landau–Widom form, whose asymptotes differ (4 against 7). Under the Riemann Hypothesis the floor is the inverse of a sampling constant for the zeros above the horizon, and a prolate-type count gives $\Phi \gtrsim \pi T^*/\log(4aT^*)$, one quarter of what is observed. Nothing in this paper proves the Riemann Hypothesis. Theorems, computations with their precision, and conjectures are distinguished throughout.

Standalone research paper. Companion to Primes, Folds, and the One Dot and to Theta Kernels, Weil Positivity, and the Mathematical Theory Atlas.

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Status conventions

A **Theorem** or **Proposition** without citation is proved in the text. A statement with a citation is taken from the literature and its hypotheses are stated. A **Computation** is a numerical result with its method, precision and cross-checks; a finite-mode minimum is a minimum over a subspace and hence an upper bound on the floor, and every such value is labelled by its mode count. A **Conjecture** is a precise statement with the evidence for it. A **Heuristic** or **Remark** makes no mathematical claim.

1 The form, the floor and the horizon

For $f \in L^2(\mathbb{R})$ real with compact support write $F = \hat{f}$, $F(t) = \int f(x)e^{itx} dx$, and $g = f \star \tilde{f}$, $g(u) = \int f(x)f(x-u) dx$, so that $\hat{g} = |F|^2$. The Weil form in the normalization used throughout is

$$Q(f) = W_\infty(g) - \sum_{n \geq 2} \Lambda(n)n^{-1/2}(g(\log n) + g(-\log n)) + (\hat{g}(i/2) + \hat{g}(-i/2)), \quad (1)$$

$$W_\infty(g) = -(\gamma + \log \pi)g(0) + 2 \int_0^\infty \frac{g(0)e^{-2x} - g(x)e^{-x/2}}{1 - e^{-2x}} dx. \quad (2)$$

By the explicit formula, $Q(f) = \sum_{\rho} \hat{g}((\rho - \frac{1}{2})/i)$ over the nontrivial zeros; on the geometric side, in Fourier form,

$$Q(f) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(t)|^2 \Psi(t) dt + \text{polar}, \quad \Psi(t) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi - \sum_n \frac{2\Lambda(n)}{\sqrt{n}} \cos(t \log n), \quad (3)$$

where the polar term is $2F(i/2)^2$ for even f and $-2P(f)^2$, $P(f) = \int f(x) \sinh(x/2) dx$, for odd f . This is the normalization of Zhu [6] and of Checkpoints 15–20 of [15]; Computation 2.1 confirms the identification numerically.

Definition 1.1. For $a > 0$ the *odd floor* is

$$\lambda(a) = \inf \{ Q(f) : f \text{ odd, } \operatorname{supp} f \subset [-a, a], \|f\|_2 = 1 \}, \quad \Phi(a) = -\log \lambda(a),$$

and the *horizon* is $T^*(a) = 2\pi e^{2a}$.

Since g is supported in $[-2a, 2a]$, only the prime powers with $\log n < 2a$ enter $Q(f)$; the *entries* are $a_n = \frac{1}{2} \log n$. The form Q itself does not depend on a , so λ is non-increasing.

Theorem 1.2 (Weil; Yoshida; see Suzuki [4]). *The Riemann Hypothesis holds if and only if $Q(f) \geq 0$ for every compactly supported smooth f ; and it suffices that $Q(f) \geq 0$ for every odd such f . Hence $\text{RH} \iff \lambda(a) \geq 0$ for all $a > 0$.*

Theorem 1.3 (Zhu [6], Theorem 6.2). *For $a = 0.8$, $8.206 \times 10^{-15} \leq \lambda(0.8) \leq 2.347 \times 10^{-14}$, certified by interval arithmetic. In particular $\lambda(a) > 0$ for $0 < a \leq 0.8$.*

The horizon is the height at which the zeros become too dense for the window. A function of exponential type a can vanish on a real sequence of upper density at most a/π (Beurling–Malliavin [10]); the zeros have density $(2\pi)^{-1} \log(t/2\pi)$ at height t , which equals a/π exactly at $t = T^*(a)$. Zhu’s abstract states the same fact as a resolution threshold: a certificate at half-width a must resolve frequencies of order $T^*(a)$, doubly exponential in a [6]. Everything below is organized around the pair $(\lambda(a), T^*(a))$.

2 Three engines

All computations use the orthonormal odd Legendre basis on the window, $f(y) = a^{-1/2} \sum_{i < K} c_i N_i P_{2i+1}(y/a)$, $N_i = \sqrt{(4i+3)/2}$, in which the matrix of (1) is assembled from the exact cross-correlations $g_{ij}(x) = \int \phi_i(u) \phi_j(u-x) du$ by Gauss–Legendre rules exact for the polynomial products. Three implementations are used, all in the repository accompanying this paper [17].

1. **rh_weil_odd.py**: mpmath, 50–60 digits, $K \leq 72$. Reproduces the floors and the failure points of [14] to 4–6 digits.
2. **rh_weil_arb.py**: ball arithmetic (python-flint/arb), $K \leq 230$, 500–800 bits, with certified Gauss–Legendre rules, $M = 2K + 8$ inner and $2K + 100$ outer nodes, and a rigorous enclosure of the lowest eigenvalue of the assembled matrix. The Legendre recurrence is evaluated with a midpoint reset at each degree, since its ball version doubles the radius per degree; the enclosure therefore covers the assembly and the eigenvalue computation but not the polynomial evaluations, which are 500-bit floating point, nor the truncation of the outer rule. It reproduces the mpmath floors to twenty digits and runs a 180-mode point in six minutes.

3. `rh_edge_fem.py`: odd-extended hat functions on a mesh of the half window graded geometrically to $10^{-12}a$ (350–490 nodes), exact cross-correlations, closed-form archimedean tails, double precision. Built to read the minimizer at the edge, where a Legendre truncation resolves only $\delta \gtrsim a/K^2$.

Computation 2.1. The 40-mode floor at $a = 0.8$ is 1.5935×10^{-14} and the 64-mode floor is 1.5659×10^{-14} , both inside Zhu’s certified interval of Theorem 1.3, which fixes the normalization and shows the truncation error at that support. At $a = 0.3$, where no prime is inside the window, the 64-mode floor 0.2225569 and the edge-FEM floor 0.2225566 agree to 1.3×10^{-6} ; at $a_3 = \frac{1}{2} \log 3$ the 64-mode profile and the FEM profile agree to 10^{-4} for $|y| \leq 0.99a$.

3 The archimedean wall and the edge law

Proposition 3.1 (decomposition on the window). *Let $J_\Gamma(s) = e^{-|s|/2}/(1 - e^{-2|s|})$ and $c_1 = -(\gamma + \log \pi) - \frac{\pi}{2} - 3 \log 2$. For f supported in $[-a, a]$,*

$$W_\infty(g_f) = c_1 \|f\|^2 + \frac{1}{2} \iint_{[-a, a]^2} J_\Gamma(u-v) |f(u) - f(v)|^2 du dv + \int_{-a}^a V_a(y) |f(y)|^2 dy, \quad V_a(y) = \int_{|v| > a} J_\Gamma(y-v) dv,$$

and $V_a(y) = \frac{1}{2} \log \frac{1}{a-|y|} + \kappa_a + o(1)$ as $|y| \rightarrow a$, with $\kappa_a = \frac{1}{2} \log 4 + \frac{\pi}{4} + \operatorname{artanh}(e^{-a}) + \arctan(e^{-a})$.

Proof. With $g(0) - g(x) = \frac{1}{2} \int |f(u) - f(u-x)|^2 du$, the integral in (2) is $\frac{1}{2} \iint_{\mathbb{R}^2} J_\Gamma(u-v) |f(u) - f(v)|^2$, and $2 \int_0^\infty (e^{-2x} - e^{-x/2})/(1 - e^{-2x}) dx = -\pi/2 - 3 \log 2$ by the digamma series $\psi(\frac{1}{4}) = -\gamma - \frac{\pi}{2} - 3 \log 2$. Splitting the double integral according to whether v lies in the window gives the three terms. Since $\int_x^\infty J_\Gamma = \operatorname{artanh}(e^{-x/2}) + \arctan(e^{-x/2})$, $V_a(y) = F(a-y) + F(a+y)$ with this F , and $\operatorname{artanh}(e^{-x/2}) = \frac{1}{2} \log(4/x) + o(1)$ gives the constant; at $a = a_3$ it is 2.6606, which the direct evaluation of V_a confirms to five digits. $\square$

The symbol of the whole-line form $\frac{1}{2} \iint J_\Gamma |f(u) - f(v)|^2$ is $\sigma(\xi) = \int J_\Gamma(s)(1 - \cos s\xi) ds = \log |\xi| + O(1)$, so on the window the archimedean form is one half of the Dirichlet form $\int \log(\xi^2) |\hat{f}|^2$ of the logarithmic Laplacian L_Δ of Chen–Weth [8] plus a bounded quadratic form; the prime terms are bounded translation operators and the polar term has rank one. The Euler–Lagrange equation of a minimizer is therefore $L_\Delta f = F$ with $F \in L^\infty$ whenever $f \in L^\infty$.

Theorem 3.2 (edge law; Chen–Weth [8], Thm. 1.11; Hernández-Santamaría–López Ríos–Saldaña [9], Thm. 1.1). *Let f be a bounded minimizer of Q over odd functions supported in $[-a, a]$. Then f is continuous on $\mathbb{R}$ and*

$$|f(y)| \leq C \left(\log \frac{1}{a-|y|} \right)^{-1/2} \quad (|y| < a),$$

for some C . In particular $f(\pm a) = 0$. The exponent $\frac{1}{2}$ is sharp for the torsion function of a small interval [9, Thm. 1.2], and a Hopf-type lemma gives the same rate from below for nonnegative supersolutions [9, Thm. 1.4].

The same exponent comes out of the Euler–Lagrange equation directly. In $t = \log(1/\delta)$, $\delta = a - y$, the equation near the edge reads $t h(t) - \frac{1}{2} \int_0^t h(\tau) d\tau = O(h) + O(1)$, the coefficient t being $\frac{1}{2}$ from the wall and $\frac{1}{2}$ from the interior kernel, and the nonlocal term coming from the long-range part of the kernel, which in the variable t is the constant $\frac{1}{2}$ on one side; its solutions are $h = Ct^{-1/2}(1 + O(1/t))$. Checkpoint 17 of [15] applied Theorem 3.2 to the forced solutions of the same operator and left open the existence of the normalized trace $\lim \sqrt{\log(1/\delta)} f(a - \delta)$; the following computation is the numerical answer for the ground state.

Computation 3.3 (the edge amplitude). On every mesh and with or without primes, $1/f(a-\delta)^2$ is linear in $t = \log(a/\delta)$ with relative residual below 10^{-4} over $14 \leq t \leq 20$; the local exponent $-d \log f/d \log t$ descends toward $\frac{1}{2}$ exactly as $\frac{1}{2}t/(t+\beta)$ does. Writing $f(a-\delta) = C(t+\beta)^{-1/2}(1+o(1))$:

form	a	λ (FEM)	C	β	mesh dependence of C
prime-free	0.3	0.2225566	1.3089	-1.49	10^{-3}
full	$a_3 = 0.5493$	1.4914×10^{-5}	0.02412	-1.87	10^{-2}

The six K -mode endpoint values of [14] at a_3 ($f_K(a_3) = 0.009471, \dots, 0.008207$ for $K = 24, \dots, 64$) are this law read at the truncation's resolution: with the fitted C, β , the effective $t_K = C^2/f_K^2 - \beta$ equals $2 \log K + 2.0$ within 0.3 for all six, with and without primes, so a degree- $2K$ expansion resolves $\delta \approx e^{-2}a/K^2$ at the edge and $f_K(a)^2 = C^2/(2 \log K + \beta')$.

4 The relay is a soft kink

At an entry a_n the new term $-2\Lambda(n)n^{-1/2}g(\log n)$ vanishes and grows with a .

Proposition 4.1 (K -mode kink). *For the K -mode form, λ_K is continuous at a_n and $\lambda'_K(a_n^+) - \lambda'_K(a_n^-) = 4\Lambda(n)n^{-1/2}f_K(a_n)^2$, where f_K is the normalized K -mode minimizer at entry.*

Proof. For $a = a_n + \varepsilon$ and odd f , $g(\log n) = \int_{a-2\varepsilon}^a f(y)f(y-\log n) dy = -\int_0^{2\varepsilon} f(a-\delta)f(a-2\varepsilon+\delta) d\delta$, which is $-2\varepsilon f(a)^2 + O(\varepsilon^2)$ for a function continuous at the edge with $f(a) \neq 0$; the K -mode minimizer is a polynomial. Hellmann–Feynman with the minimizer frozen gives the jump. $\square$

Computation 4.2. With 24 modes at the entry of 3, the measured jump is 0.948 of the prediction at step 5×10^{-4} and 0.9998 at step 10^{-5} [17]; the deficit at coarser steps is the reorganization of the minimizer within the crossover scale gap/ $\|dT/da\| \approx 3 \times 10^{-5}$.

For the exact form, Theorem 3.2 gives $f(a_n) = 0$ and the proposition's jump is a quantity of the truncation. What replaces it follows from Computation 3.3: with $f(a-\delta) = C(t+\beta)^{-1/2}$,

$$-2\Lambda(n)n^{-1/2}g(\log n) = \frac{4\Lambda(n)C^2}{\sqrt{n}} \frac{\varepsilon}{\log(1/\varepsilon) + \beta} (1 + o(1)), \quad (4)$$

so λ' is continuous across the entry and its increment at scale ε is $4\Lambda(n)n^{-1/2}C^2/(\log(1/\varepsilon) + \beta)$: a soft kink, vanishing only logarithmically.

Computation 4.3 (the soft kink). On the edge FEM at the entry of 3 ($C = 0.0241$, $\beta = -1.93$), with $D(\varepsilon) = [\lambda(a_3 + \varepsilon) - 2\lambda(a_3) + \lambda(a_3 - \varepsilon)]/\varepsilon$, all entries $\times 10^{-4}$:

ε	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}
measured $D(\varepsilon)$	2.45	1.90	1.48	1.21	1.01
first-order energy on the a_3 minimizer	3.42	2.13	1.59	1.27	1.06
law (4)	2.96	2.02	1.54	1.24	1.04

The increment falls by 22% per decade as $1/(\log(1/\varepsilon) + \beta)$ does and agrees with the law to 2–6% from $\varepsilon = 10^{-4}$ down; at 10^{-3} the smooth curvature $\lambda''\varepsilon \approx 0.9 \times 10^{-4}$ is still inside D . The 24-mode form gives the constant 2.28×10^{-4} .

At the resolution of the grids of Section 6 (steps 0.004 to 0.025) the entries beyond $n = 3$ leave no visible trace in λ' . The exact derivative resolves what those steps average over.

Computation 4.4 (the relay resolved). `rh_dilation.py` at $a_n + \text{offset}$ for the first five entries; entries are Φ'/T^* (data in `relay_scan.csv`):

n	a_n	K	-0.01	-0.003	-0.001	-0.0001	0.0001	0.001	0.01
2	0.3466	48	2.42	2.96	3.16	3.27	2.80	2.59	2.30
3	0.5493	48	2.31	3.02	3.65	4.08	3.41	3.17	3.08
4	0.6931	64	3.61	3.32	3.70	4.08	3.73	3.63	3.51
5	0.8047	80	3.96	2.98	3.32	4.32	3.63	3.48	3.51
7	0.9730	96	3.18	3.66	3.06	4.12	3.59	3.54	3.84

Every entry shows the same shape: a rise of 30–45% over the last 0.003 before the entry, a drop across it at scale 10^{-4} (14%, 16%, 9%, 16%, 13% for $n = 2, 3, 4, 5, 7$; at this scale the K -mode kink of Proposition 4.1 and the soft kink (4) nearly coincide, since $1/K^2 \approx 10^{-4}$), and a slow relaxation after. The rise is the form without the entering prime power heading toward its failure point a_{fail} , the deleted-form failure of [14] (0.5574 for $n = 3$): λ falls roughly linearly there ($\lambda' \approx -1.14 \times 10^{-3}$ at $a = 0.549$, reaching zero at 0.562 by linear extrapolation) while λ' barely moves, so $\Phi' = -\lambda'/\lambda$ grows like $1/(a_{\text{fail}} - a)$ until the entry rescues the floor. Each value was checked against a finite difference of step 2×10^{-4} at $a = 0.548$ – 0.549 to four digits. The peaks at the entries are 3.27, 4.08, 4.08, 4.32, 4.12 and the dips between them reach 2.3; the coarse grids average this sawtooth into the plateau of Computation 6.1.

The relay is therefore the sawtooth described in [14], measured exactly: not a kick at the entry but a pole-like rise into it, cut off by the entering prime power, whose first-order action at the entry is the soft kink. The crossover scale $\text{gap}/\|dT/da\|$ (3×10^{-5} at a_3 , 10^{-9} at $a = 0.7$) is where the minimizer reorganizes; the sawtooth lives three to four decades above it.

5 The sliding-window identity

Theorem 5.1 (sliding-window identity). *Let a lie strictly between two consecutive entries and let f be a minimizer at support a (exact, or the K -mode minimizer, in which case λ means λ_K). Then*

$$a \lambda'(a) = - \left[D_\infty(f) + D_P(f) + 2P(f)^2 + \frac{2P(f)}{a} M(f) \right], \quad (5)$$

where, with $F = \hat{f}$ and $g = g_f$,

$$D_\infty(f) = \frac{1}{2\pi} \int |F(t)|^2 t \partial_t \text{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) dt, \quad D_P(f) = - \sum_{\log n < 2a} \frac{2\Lambda(n)}{\sqrt{n}} \log n g'(\log n),$$

$$P(f) = \int f(x) \sinh(x/2) dx \text{ and } M(f) = \int f(x) x \cosh(x/2) dx.$$

Proof. For the dilation family $f_b(x) = b^{-1/2} f(x/b)$, b near 1, one has $\hat{f}_b(t) = \sqrt{b} F(bt)$ exactly, jumps included. Between entries the prime set in Ψ of (3) is fixed, and the substitution $s = bt$ gives $\frac{d}{db} \frac{1}{2\pi} \int |\hat{f}_b|^2 \Psi = -\frac{1}{b} \cdot \frac{1}{2\pi} \int |\hat{f}_b(t)|^2 t \Psi'(t) dt$. The symbol $t \Psi'(t)$ is $t \partial_t \text{Re} \psi(\frac{1}{4} + \frac{it}{2}) + \sum 2\Lambda(n) n^{-1/2} (t \log n) \sin(t \log n)$, and $\frac{1}{2\pi} \int |F|^2 t \sin(tu) dt = -g'(u)$ for real f , which gives D_∞ and D_P . For the polar term, $P(f_b) = \sqrt{b} \int f(y) \sinh(by/2) dy$ gives $\frac{d}{db} P(f_b)|_{b=1} = \frac{1}{2} P(f) + \frac{1}{2} M(f)$,

so $\frac{d}{db}(-2P(f_b)^2) = -2P^2 - 2PM$. The K -mode basis of Section 2 is a dilation family, so the matrix $Q_K(a)$ is Q on $\{f_{a/a_0}\}$ and Hellmann–Feynman for the simple lowest eigenvalue gives $\lambda'_K(a) = \langle c, Q'_K(a)c \rangle$, which is the right side divided by a ; the exact form is the same computation on the exact minimizer, whose lowest eigenvalue is simple in the range considered (Computation 6.1). $\square$

Since $t \partial_t \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) \rightarrow 1$, $D_\infty(f) = \|f\|^2 + O(\int |F|^2 t^{-2})$; on a minimizer $\|f\| = 1$ while λ is 10^{-90} at $a = 1.5$, so (5) forces

$$D_P(f) = -D_\infty(f) - 2P^2 - \frac{2P}{a}M + O(aT^*\lambda) : \quad (6)$$

the Euler product, differentiated and applied to the derivative of the minimizer’s autocorrelation at the prime-power positions, cancels the archimedean dilation form to ninety digits. Together with the explicit formula on the same function, $\sum 2\Lambda(n)n^{-1/2}g(\log n) = W_\infty(g) - 2P^2 - \lambda$, both the values and the log-weighted derivatives of g_f at the prime powers are pinned by the archimedean side.

Computation 5.2 (the identity on the minimizer). `rh_dilation.py` differentiates the three parts of $Q_K(a)$ by central differences with step 10^{-30} in ball arithmetic and contracts them with the eigenvector; every entry is a ball with the radius shown.

a	K	λ_K	$a\langle c, A'c \rangle$	$a\langle c, P'c \rangle$	$a\langle c, S'c \rangle$	sum = $a\lambda'$	radius
0.6	48	5.9659×10^{-7}	-1.014313	+1.086974	-0.072687	-2.493408×10^{-5}	4×10^{-20}
0.8	64	1.5659×10^{-14}	-1.016605	+1.103966	-0.087361	$-1.295854 \times 10^{-12}$	4×10^{-27}
1.0	80	1.4820×10^{-26}	-1.018057	+1.114966	-0.096909	$-2.516852 \times 10^{-24}$	3×10^{-39}
1.25	140	1.5442×10^{-50}	-1.019178	+1.123585	-0.104407	$-1.233712 \times 10^{-48}$	2×10^{-63}

Here A, P, S are the archimedean, prime and polar matrices, so $a\langle c, A'c \rangle = -D_\infty$ and $a\langle c, P'c \rangle = -D_P$. The archimedean dilation form sits at 1.014–1.019, the excess over $\|f\|^2$ being the finite- t correction; the prime dilation form grows with the primes present and cancels it; the sum equals $-a\Phi'\lambda$ with $\Phi'/T^* = 3.339, 3.324, 3.658, 3.774$, matching the grids of Section 6.

5.1 The derivative is a property of the ground state

Is there an inequality $-Q' \leq cT^*Q$ between the two forms, which would give the derivative bound for every f at once? The pencil $(-Q', Q)$ answers this.

Computation 5.3 (the pencil). Largest and smallest eigenvalues of $Q^{-1}(-Q')$ in the K -mode space, divided by T^* :

a	minimizer direction	largest	second, third	smallest
0.6	3.34	74.7	4.26, 3.69	-70.5
0.8	3.32	1.46×10^5	332, 6.08	-1.46×10^5
1.0	3.66	6.2×10^{10}	$3.1 \times 10^7, 8.2 \times 10^4$	-6.2×10^{10}

The extreme eigenvalues come in $\pm$ pairs and grow explosively; they are near-null directions of Q whose dilation derivative is enormous relative to their value, the reorganization directions of the minimizer, and their size is the inverse of the crossover scale (10^{-7} at $a = 0.8$, 10^{-13} at $a = 1$). The ground state is neither extreme; it is a special interior direction in which the dilation derivative is

anomalously small, by ten orders at $a = 1$. Hence the derivative bound of Section 7 is a statement about the ground-state eigenvector alone, $\lambda' = \langle f_0, Q' f_0 \rangle$, and cannot be an inequality between forms. A proof must pass through the Euler–Lagrange equation of the minimizer.

6 The floor to height 130

Computation 6.1 (the band). Φ' is computed from pairs $a \pm h$ ($h = 0.0125$ on a uniform grid with 40 modes on $[0.30, 1.20]$, $h = 0.0125$ with 72 modes on $[0.475, 1.05]$, $h = 0.005$ with the ball-arithmetic engine at 110–230 modes on $[1.055, 1.505]$), never across an entry. Convergence: the 40-mode floor agrees with 72 modes to 5% up to $a = 0.925$ and 26% at 1.0; the 140-mode floor agrees with 180 modes to 2% at 1.25; at 1.5, 180 against 230 modes differ by 39% in λ and 4% in the ratio, more modes raising it, so the last ratio is a lower bound. Every eigenvalue beyond $a = 1$ carries a rigorous enclosure of the assembled matrix’s lowest eigenvalue, radius $\approx 10^{-143}$ at 500 bits and 10^{-140} at 800 bits.

a	K	T^*	λ	Φ'	Φ'/T^*
0.500	72	17.1	1.938×10^{-4}	55.2	3.23
0.650	72	23.1	1.837×10^{-8}	74.8	3.24
0.775	72	29.6	2.105×10^{-13}	107.5	3.63
0.900	72	38.0	7.176×10^{-20}	138.5	3.64
1.000	72	46.4	1.491×10^{-26}	172.6	3.72
1.060	110	52.4	2.97×10^{-31}	195.8	3.74
1.150	110	62.7	1.04×10^{-39}	240.3	3.83
1.250	180	76.5	3.59×10^{-51}	289.7	3.79
1.330	140	89.8	3.58×10^{-62}	339.2	3.78
1.400	180	103.3	1.73×10^{-73}	398.4	3.86
1.450	180	114.2	1.22×10^{-82}	443.5	3.88
1.500	230	126.2	2.97×10^{-93}	491.1	3.89

Here λ at a centre is the geometric mean of the pair. Means of Φ'/T^* : 3.23 on $[0.5, 0.7]$, 3.61 on $[0.85, 1.05]$, 3.74 on $[1.06, 1.25]$, 3.85 on $[1.33, 1.5]$. Least squares of Φ'/T^* against a for $a \geq 0.85$, by rms relative residual: $\Phi' = cT^*$ with $c = 3.72$, 3.4%; Zhu’s form $\Phi \simeq C N(T^*)/\log N(T^*)$, whose ratio is $c(2a - 1)/(2a + \log(2a - 1))$, 3.0% with $c = 7.2$; $\Phi' = cT^* \log T^*$, 7.8%; the raw count $\Phi \propto N(T^*)$, ratio $\propto 4a - 1$, 21%. The ratio λ_1/λ_0 grows without interruption (5.7 at $a = 0.31$, 10^6 at $a = 1$), so the ground state stays simple across every entry.

At step 0.004 the entries beyond $n = 3$ show no net drop in Φ' , because the difference quotient across an entry averages the steep rise before it against the drop at it; Computation 4.4 resolves both. Over the fine scan, Φ'/T^* lies in $[2.3, 4.3]$, and the supremum over all measured points, the quantity Conjecture 7.1 bounds, is 4.32, at the entry of 5.

7 The conjectures

Conjecture 7.1 (bounded relay). *There are $a_0 > 0$ and $c > 0$ such that $\Phi'(a) \leq cT^*(a)$ for all $a \geq a_0$.*

Proposition 7.2. *Conjecture 7.1 with any $a_0 \leq 0.8$ implies the Riemann Hypothesis.*

Proof. Φ is locally Lipschitz and, by Theorem 3.2 and (4), Φ' is continuous across every entry, so the inequality integrates: $\lambda(a) \geq \lambda(a_0) \exp(-c(T^*(a) - T^*(a_0))) > 0$ for $a \geq a_0$, using Theorem 1.3 at $a_0 = 0.8$. Theorem 1.2 concludes. $\square$

Conjecture 7.3 (decay law). $\Phi'(a)/T^*(a)$ is bounded above and below by positive constants for $a \geq a_0$. In its sharp form, $\Phi'(a)/T^*(a) \rightarrow c_0$, i.e. $\lambda(a) = \exp(-(c_0/2 + o(1))2\pi e^{2a})$.

Computation 6.1 supports the weak form from height 17 to 130 and leaves the sharp constant undetermined: the ratio rises by shrinking steps, the pure horizon law and Zhu's Landau–Widom form fit equally well, and they disagree on the asymptote ($c_0 \approx 4$ against $c_0 \approx 7$, the second reached only far beyond any computable height). The raw phase-space count is excluded and the logarithmic law is disfavoured two to one. Conjecture 7.1 is the one that implies the Riemann Hypothesis; the constant is not its content.

By Theorem 5.1, Conjecture 7.1 is the inequality

$$D_\infty(f) + D_P(f) + 2P(f)^2 + \frac{2P(f)}{a}M(f) \leq caT^*(a)Q(f) \quad \text{on the minimizer } f, \quad (7)$$

and by Computation 5.3 it holds only there.

Remark 7.4 (what a proof cannot be). Positivity of Q on a window is destroyed by displacing $\log 2$ by 3×10^{-14} at $a = 0.8$ and by $\exp(-cT^*(a))$ in general [14, 17], so no statement about the primes that is invariant under such displacements can imply it; and Checkpoint 20 of [15] shows that bounding the reflected prime terms separately from the polar term yields $-\infty$ on translated tests. A proof of Conjecture 7.1 must therefore use the exact multiplicative structure of Λ , and by Computation 5.3 it must use that f minimizes Q .

7.1 Under the Riemann Hypothesis

If all zeros are real, $Q(f) = \sum_\gamma |F(\gamma)|^2$ and the same dilation computation gives $a\lambda' = \lambda + \sum_\gamma \gamma \partial_t |F|^2(\gamma)$, so Conjecture 7.1 reads $\sum_\gamma \gamma \partial_t |F|^2(\gamma) \geq -(1 + caT^*) \sum_\gamma |F(\gamma)|^2$: the minimizer is small at the zeros to the same order in its derivative as in its value. And

$$\Phi(a) = -\log \inf\{\sum_\gamma |F(\gamma)|^2 : F \in \text{PW}_a, \|f\| = 1\}$$

is the logarithm of a sampling constant for the zeros above the horizon.

Heuristic 7.5 (prolate lower bound). The number of zeros in $[-T^*, T^*]$ is $2N(T^*) \approx (T^*/\pi)(2a - 1)$, short of the Shannon number $2aT^*/\pi$ of $[-a, a] \times [-T^*, T^*]$ by T^*/π . By Landau–Pollak–Slepian [11] and the asymptotics of the prolate spectrum [12], a function in PW_a can vanish on all of them and keep all but $\exp(-\pi T^*/\log(4aT^*))$ of its energy below the horizon; under RH this bounds λ from above, i.e. $\Phi \gtrsim \pi T^*/\log(4aT^*) \approx 0.5 T^*$ at $a = 1$, against the observed $1.85 T^*$. The zeros are about four times more expensive a constraint set than the count alone predicts. Conjecture 7.1 is the matching upper bound.

8 The boundary law

The derivative of the floor is a boundary quantity. Since Q does not depend on a , $\lambda(a)$ is the infimum of a fixed form over the growing family $L^2(-a, a)$, and its derivative is a domain variation. For the Dirichlet fractional Laplacian, Hadamard's formula expresses such a derivative through the boundary trace u/dist^s of the eigenfunction [13]; for the logarithmic Laplacian the boundary trace is the amplitude C of Theorem 3.2 and Computation 3.3, and a Pohozaev identity with the local

boundary term $\int_{\partial\Omega} u^2 \ln(\delta^{-2}) (x \cdot \nu)$ has been announced but, to our knowledge, not established (a 2024 preprint claiming it was withdrawn). We therefore state the law as a conjecture and give the evidence.

Conjecture 8.1 (boundary law). *For a not an entry, with $C_{\pm}(a) = \lim_{\delta \rightarrow 0} \sqrt{\log(1/\delta)} |f(\pm(a-\delta))|$ the edge amplitudes of the minimizer,*

$$\lambda'(a) = -(C_+(a)^2 + C_-(a)^2) = -2C(a)^2.$$

Equivalently, with Theorem 5.1, the Weil form on a window satisfies the Pohozaev-type identity $D_{\infty}(f) + D_P(f) + 2P(f)^2 + \frac{2P(f)}{a}M(f) = 2aC(a)^2$ on its minimizer.

Computation 8.2 (the boundary law). C from the edge FEM (two-stage mesh to $10^{-12}a$, fit of $1/f^2$ against $\log(a/\delta)$ on $14 \leq t \leq 20$, residual $\leq 5 \times 10^{-5}$) and λ' from `rh_dilation.py` (exact, ball arithmetic, 48 modes):

a	primes	λ (FEM)	C	β	λ'	$\kappa = -\lambda'/2C^2$
0.3	no	+0.2225566	1.30891	-1.49	-3.42881	1.0007
0.4	no	-0.0780502	1.15425	-1.48	-2.66640	1.0007
0.5	no	-0.3222425	1.06222	-1.45	-2.25815	1.0007
0.6	no	-0.5358860	1.00932	-1.42	-2.03880	1.0007
0.4	yes	1.470510×10^{-2}	0.48266	-1.57	-0.465947	1.0001
0.45	yes	2.404578×10^{-3}	0.22494	-1.67	-0.101283	1.0009
0.5	yes	1.936801×10^{-4}	0.07493	-1.78	-1.123556×10^{-2}	1.0005
0.549	yes	1.491447×10^{-5}	0.02412	-1.87	-1.143529×10^{-3}	0.983

Seven cases give $\kappa = 1$ to better than 10^{-3} , across supports where the prime-free floor is positive and negative and with the primes present; the common offset 7×10^{-4} is the mesh bias of C . The last case measures C at $a_3 = 0.54931$ and λ' at 0.549, in the region where Φ' changes by 6% per 0.001 (Section 6), which accounts for its 1.7%.

An exactly solvable check: for one half of the Dirichlet form of L_{Δ} on $(-a, a)$, the scaling $L_{\Delta}(u(\cdot/a)) = (L_{\Delta}u)(\cdot/a) + 2\log a u(\cdot/a)$ gives $\lambda(a) = \lambda(1) - \log a$, so the law predicts $C(a)^2 = 1/(2a)$; the prime-free amplitudes above give $2aC^2 = 1.028, 1.066, 1.129, 1.223$ at $a = 0.3, \dots, 0.6$, and the exact derivatives give $-a\lambda' = 1.029, 1.067, 1.129, 1.223$: the departures from 1 are the bounded corrections (the polar term and $\operatorname{Re} \psi - \log |t|$), identical on both sides.

Corollary 8.3 (of Conjecture 8.1). $\Phi'(a) = 2C(a)^2/\lambda(a)$, and Conjecture 7.1 is equivalent to

$$C(a)^2 \leq \frac{\varepsilon}{2} T^*(a) \lambda(a) \quad (a \geq a_0) :$$

the edge amplitude of the minimizer is bounded by its energy times the horizon.

This is the form in which the conjecture is a statement about one function at one place. The relay of Section 4 reads: as a approaches an entry from below, the form without the entering prime power is heading to indefiniteness at a_{fail} (the deleted-form failure points of [14]), λ falls roughly linearly while C stays of the same order, and $\Phi' = 2C^2/\lambda$ rises like $1/(a_{\text{fail}} - a)$ until the entry rescues the floor; Computation 4.4 records the rise at every entry (Φ'/T^* from 3.20 at $a = 0.547$ to 4.08 at $a = 0.5492$, verified both by the identity and by finite differences of step 2×10^{-4}). Under the Riemann Hypothesis, $C(a)$ is the boundary trace of the band-limited function that best hides from the zeros above the horizon, and the corollary says that this trace, squared, never exceeds a horizon-multiple of the residual energy at the zeros.

9 Relation to the literature

Suzuki [4] organizes the Weil form on $[-a, a]$ as a family of self-adjoint operators and conjectures that an operator with the zeros as spectrum is their limit as $a \rightarrow \infty$; Conjecture 7.1 is a differential inequality for the ground state of that family. Zhu [6] certifies both parity sectors at $a = 0.8$, proposes the Landau–Widom decay law from his own upper bounds on $0.5 \leq a \leq 2$, and states the resolution threshold; his exploratory 950-mode value at $a = 1.19$ lies between 10^{-48} and 10^{-46} . Connes–Consani [7] treat the range $a \leq \frac{1}{2} \log 2$ where only the archimedean place acts; the wall of Proposition 3.1 is that place acting alone at the edge for every a . The atlas [15] contains an exact-rational certificate of the odd floor to $a = 3/8$ with constant $1/200$, inside Zhu’s range, and the first application of Theorem 3.2 to this operator. The folds paper [14] computed the floors to $a = 1.1$, the deleted-form failure points, and the symbol’s negative set, and stated the relay as a sequence of kicks, which Sections 4 and 6 replace by the continuous reorganization measured here.

Reproducibility

All scripts, data files and the memo from which this paper is drawn are in [17]: `rh_weil_odd.py`, `rh_weil_arb.py`, `rh_edge_fem.py`, `rh_soft_kink.py`, `rh_floor_grid.py`, `rh_arb_grid.py`, `rh_dilation.py`, `rh_hadamard.py`, and the data `floor_grid_K40.csv`, `floor_grid_K72.csv`, `floor_grid_arb.csv`, `boundary_law.csv`, `relay_scan.csv`.

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